Scanned Code Report

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Code Info

Auditor Scan

Scan ID 4	Date June 08, 2026
Organization CMTA	Repository SnapshotEngine
Branch dev	Commit Hash dec31b27...cdcaeb7d

Contracts in scope

contracts/base/CMTATInternalSnapshotBase.sol
contracts/base/SnapshotEngineBase.sol
contracts/deployment/CMTATStandaloneInternalSnapshot.sol
contracts/deployment/CMTATUpgradeableInternalSnapshot.sol
contracts/deployment/SnapshotEngine.sol
contracts/deployment/SnapshotEngineOwnable2Step.sol
contracts/library/Errors.sol
contracts/library/SnapshotBase.sol
contracts/library/SnapshotStateInternal.sol
contracts/interface/IERC20SnapshotCompatible.sol
contracts/interface/ISnapshotBase.sol
contracts/interface/ISnapshotScheduler.sol
contracts/interface/ISnapshotState.sol
contracts/modules/SnapshotSchedulerModule.sol
contracts/modules/SnapshotStateModule.sol
contracts/modules/SnapshotUpdateModule.sol
contracts/modules/VersionModule.sol

Code Statistics

Findings 1	Contracts Scanned 17	Lines of Code 1350

Findings Summary

Total Findings

- High Risk (0)
- Medium Risk (0)
- Low Risk (0)
- Info (1)
- Best Practices (0)

Code Summary

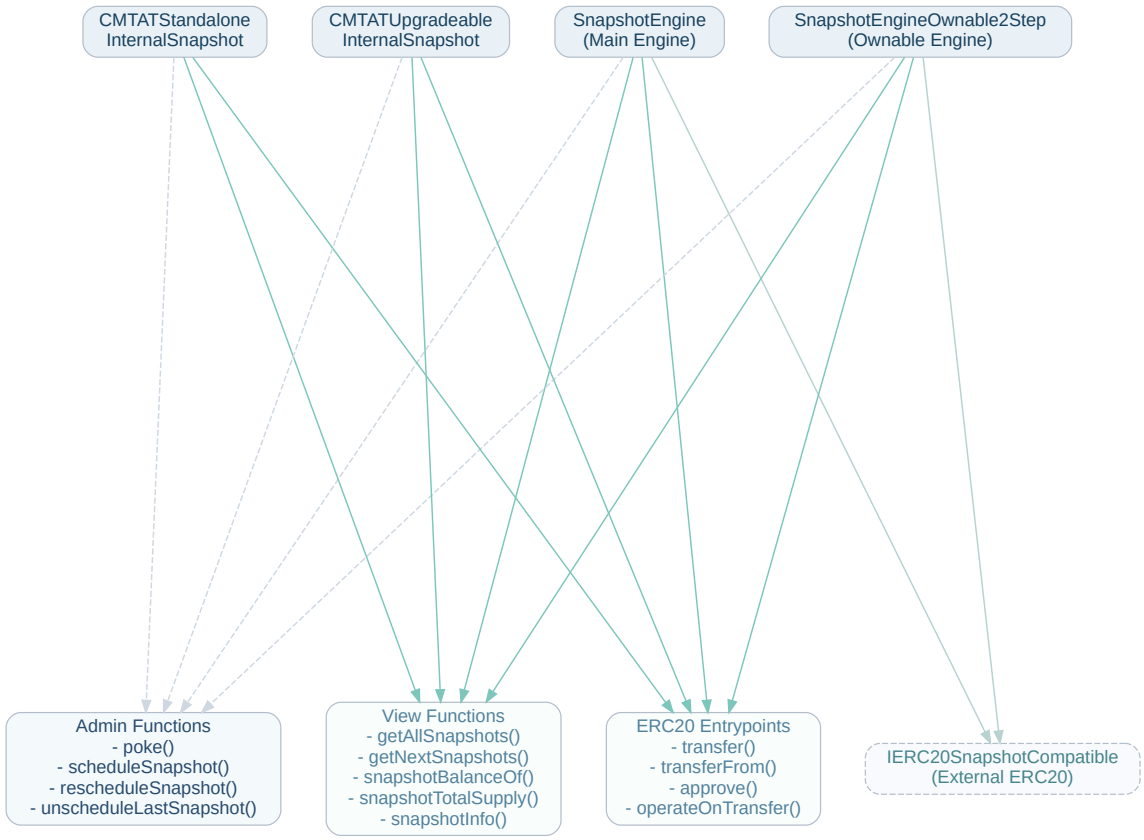
The protocol implements a snapshotting engine designed for ERC-20 tokens (specifically integrating with the CMTAT standard). It allows for scheduling, modifying, and canceling snapshots of token balances and total supply at precise historical timestamps instead of block numbers or discrete IDs.

The system consists of modules to handle snapshot state queries, scheduling operations, and balance updates on token transfers, mints, and burns. It can be deployed internally as part of a standalone token contract (e.g., `CMTATStandaloneInternalSnapshot` or `CMTATUpgradeableInternalSnapshot`) or externally bound to a compatible ERC-20 token using deployment wrappers like `SnapshotEngine` or `SnapshotEngineOwnable2Step`.

Main Entry Points and Actors

- `transfer(to, value)`: Called by Token Holders to transfer tokens.
- `transferFrom(from, to, value)`: Called by Spenders to transfer tokens on behalf of a Token Holder.
- `approve(spender, value)`: Called by Token Holders to approve a spender.
- `poke()`: Called by Snapshot Managers (SNAPSHOOTER_ROLE) to manually materialize the latest eligible scheduled snapshot.
- `scheduleSnapshot(time)`: Called by Snapshot Managers to schedule a snapshot at a specific time.
- `scheduleSnapshotNotOptimized(time)`: Called by Snapshot Managers to schedule a snapshot without optimization assumptions.
- `rescheduleSnapshot(oldTime, newTime)`: Called by Snapshot Managers to change the time of a scheduled snapshot.
- `unscheduleLastSnapshot(time)`: Called by Snapshot Managers to cancel the last scheduled snapshot.
- `unscheduleSnapshotNotOptimized(time)`: Called by Snapshot Managers to cancel a scheduled snapshot without optimization assumptions.
- `operateOnTransfer(from, to, balanceFrom, balanceTo, totalSupply)`: Called by the bound ERC-20 token contract to update snapshot states during token transfers (in external engine deployments).

Code Diagram



1 of 1 Findings

contracts/library/SnapshotBase.sol contracts/modules/SnapshotUpdateModule.sol contracts/base/SnapshotEngineBase.sol contracts/base/CMTATInternalSnapshotBase.sol

Unbounded scan of overdue scheduled snapshots can make transfer hooks and poke run out of gas

Info

The transfer hook always attempts to advance the snapshot pointer before updating account snapshots. `SnapshotUpdateModule._snapshotUpdate()` calls `SnapshotBase._setCurrentSnapshot()`, and `_setCurrentSnapshot()` calls `_findScheduledMostRecentPastSnapshot()`. That helper linearly scans `_scheduledSnapshots` from `_currentSnapshotIndex` until it reaches the first future timestamp:

`function _snapshotUpdate(...) internal virtual {
SnapshotBase._setCurrentSnapshot();
...
}

for (uint256 i = $._currentSnapshotIndex; i < currentArraySize; ++i) {
if ($._scheduledSnapshots[i] <= block.timestamp) {
mostRecent = $._scheduledSnapshots[i];
index = i;
} else {
break;
}
}`

If a large backlog of scheduled snapshots becomes overdue while no transfer or `poke()` advances the pointer, every later transfer, mint, burn, or `poke()` must scan the entire overdue suffix in one transaction. When the backlog is large enough to exceed the block gas limit, the call fails before `_currentSnapshotIndex` is updated, so subsequent attempts start from the same stale index and fail again. The due snapshots also cannot be removed through the unscheduling functions because `_checkTimeSnapshotAlreadyDone(time)` reverts once `time <= block.timestamp`. In the internal CMTAT variant this blocks `_update()` before the ERC-20 balance update, and in the external-engine variant it makes the bound token's `operateOnTransfer()` call fail and roll back the token transfer.

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